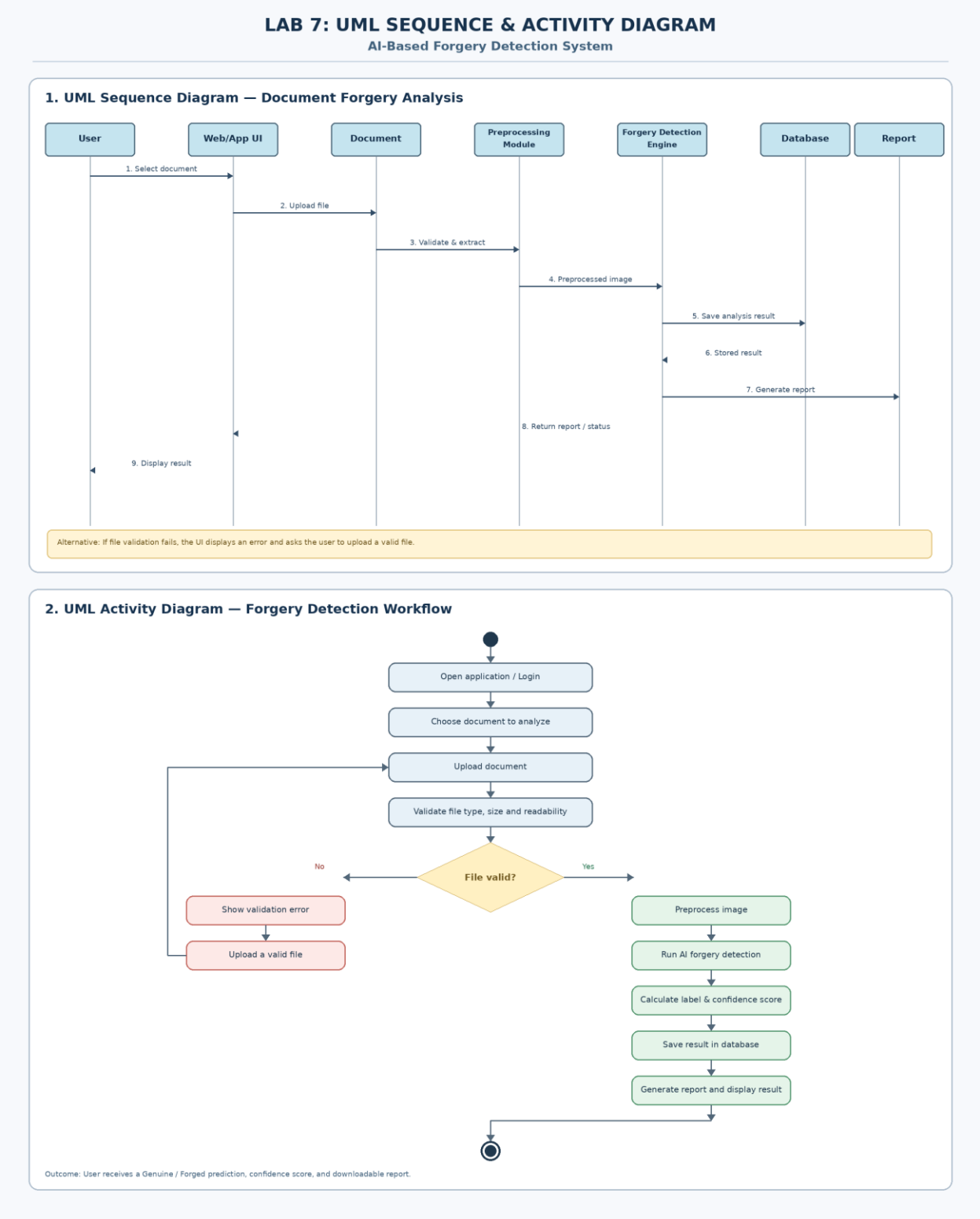
Lab 7: UML Sequence and Activity Diagrams



AI-Based Forgery Detection System

1. Introduction

UML behavioural diagrams show how a system works over time, in contrast to a class diagram, which shows only its structure. This lab uses two of them for the AI-based forgery detection system. The **sequence diagram** shows which components exchange which messages, and in what order, when a document is analysed. The **activity diagram** shows the overall workflow as a series of actions and decisions, from opening the application to receiving the result. Used together, they describe the same process from two points of view: who talks to whom, and what happens step by step.

2. Sequence Diagram: Document Forgery Analysis

2.1 Participants (Lifelines)

The diagram has seven participants, each drawn as a box at the top with a dashed vertical line, called a lifeline, running downwards.

Component	Role in the scenario
User	Starts the process by choosing a document and finally sees the result.
Web/App UI	The interface between the user and the system; sends the file and shows the output or errors.
Document	Receives the uploaded file and checks that it is valid.
Preprocessing Module	Cleans and prepares the image so the model can analyse it.
Forgery Detection Engine	The AI model that decides whether the document is genuine or forged.
Database	Stores the analysis result for later retrieval and history.
Report	Produces the final report from the stored result.

2.2 Message Flow

Time runs from top to bottom, and the numbered arrows show the order of the messages:

1. **Select document:** the user chooses a file in the interface.
2. **Upload file:** the UI sends the file to the Document component.
3. **Validate & extract:** the document is validated and its content extracted, then passed to the Preprocessing Module.
4. **Preprocessed image:** the prepared image is sent to the Forgery Detection Engine.
5. **Save analysis result:** the engine sends its result to the Database.
6. **Stored result:** the Database returns a confirmation to the engine (a return message, shown with a leftward arrow).

7. **Generate report:** the engine asks the Report component to build a report.
8. **Return report / status:** the report or status goes back to the UI.
9. **Display result:** the UI shows the outcome to the user.

Most messages are synchronous calls that move left to right, while the return messages (6, 8 and 9) travel back towards the caller.

2.3 Alternative Flow

The note at the bottom of the diagram covers the failure case: if file validation fails, the UI displays an error and asks the user to upload a valid file. This keeps the main scenario simple while still recording how errors are handled.

3. Activity Diagram: Forgery Detection Workflow

3.1 Notation Used

The activity diagram uses the following standard elements:

- **Initial node:** the solid black circle at the top, where the workflow starts.
- **Action nodes:** the rounded rectangles, each describing one step.
- **Control flow:** the arrows that connect the steps in order.
- **Decision node:** the diamond labelled "File valid?", with Yes and No branches.
- **Final node:** the circle with a ring at the bottom, where the workflow ends.

3.2 Workflow

The user first opens the application and logs in, chooses a document, uploads it, and the system validates its type, size and readability. The decision node then splits the flow in two:

- **No (invalid file):** the system shows a validation error and the user uploads a valid file. A flow arrow then returns to the "Upload document" action, forming a loop that repeats until the file is acceptable.
- **Yes (valid file):** the system preprocesses the image, runs the AI forgery detection, calculates the label and confidence score, saves the result in the database, and generates a report and displays the result. The flow then reaches the final node.

The outcome noted on the diagram is that the user receives a Genuine or Forged prediction, a confidence score and a downloadable report.

4. Comparing the Two Diagrams

Both diagrams describe the same process and agree on its main stages: upload, validation, preprocessing, detection, saving, and reporting. They differ in emphasis.

Aspect	Sequence diagram	Activity diagram
Focus	Interaction between components	Order of actions and decisions

Aspect	Sequence diagram	Activity diagram
Shows	Messages, lifelines, return values	Steps, branches, loops
Error handling	Written as a note on the alternative flow	Drawn as a decision with a retry loop
Best used for	Designing component responsibilities and interfaces	Understanding and explaining the overall workflow

5. Conclusion

The sequence diagram identifies the components involved and the messages they exchange, while the activity diagram gives the end-to-end workflow including the validation loop. Together they show that the system is organised as a clear pipeline, in which each component has one job: the document is validated, preprocessed, analysed by the AI engine, stored, reported and displayed. They also make the design easy to check, since a developer or tester can follow the same path from start to finish, and they match the class diagram, in which the same components appear as classes.